

Egress, endpoint and the partner laboratory

Day 4 - Afternoon

Running scenario: Thara Hospitals patient portal pilot

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CONCEPT 1



Timed builds

15 min

Three rules for this afternoon

- The NAT gateway bills **from creation**: about USD 0.056 an hour in Mumbai
- It is created, used and deleted inside segment 1
- The Elastic IP bills while it is yours: release it in the same segment
- Everything today is tagged `Day=4`, including what you delete

Your ledger records when the gateway was created and when it was deleted.

COMMON MISTAKE

Deleting the NAT gateway releases the Elastic IP.

It does not. The address bills until you release it.

What comes back, and what has no name

- The image holds no credentials, so step 8 types the Day 2 key into `thara-web-1` once more
- The key is switched on for the test and off again in teardown
- The gateway, the address, the endpoint and the peering carry the three tags and no name
- The partner's network is `partnerlab-vpc`, and its server `partnerlab-1`

COMMON MISTAKE

One side's route is enough for peering.

Both sides need a route. The reply needs its own.

Egress, endpoint and the partner laboratory

You build:

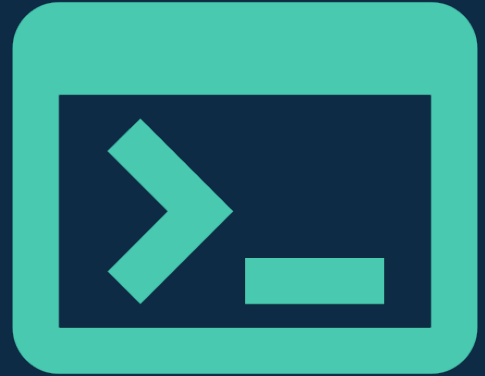
7

8

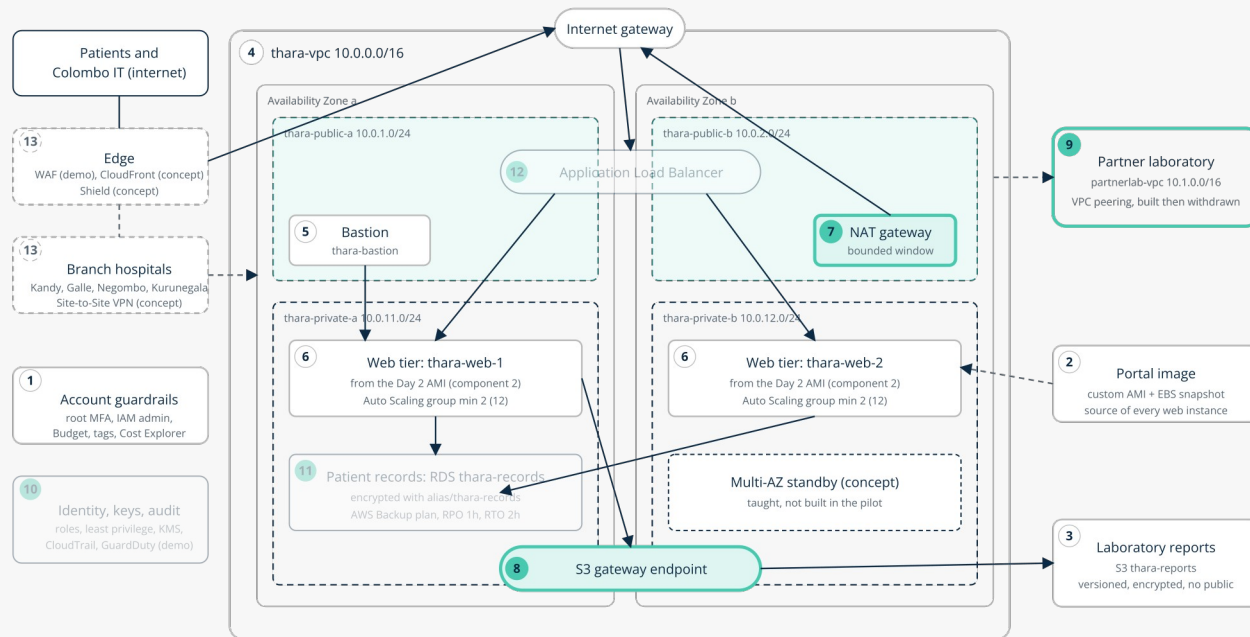
9

Estimated credit use: about USD 1.50

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The Thara architecture



SEGMENT 1 - 50 MIN



NAT gateway create, use, delete

Cost: USD 0.056 per hour plus data, and USD 0.005 for IPv4

Start, reopen the door, check the baseline

1

Confirm the region is `ap-south-1` and that you are the admin user, not root.

Expected: Mumbai, and `admin-<student id>`, in the top bar.

2

Start `thara-bastion` and `thara-web-1`. Set the SSH source of `thara-bastion-sg` to My IP again. Connect, and hop to `thara-web-1`.

Expected: A prompt of the form `ec2-user@ip-10-0-11-...`

3

On `thara-web-1`, try the internet. Screenshot the routes of `thara-private-rt`.

Expected: A timeout. One route, to local.

```
$ curl -m 10 -I https://aws.amazon.com
```

Create the NAT gateway and route to it



4

Allocate an Elastic IP. Create a NAT gateway: Zonal, subnet `thara-public-a`, Public, with that address. The three tags.

Expected: Pending, then Available within a few minutes.

5

In `thara-private-rt`, add 0.0.0.0/0 with the NAT gateway as its target.

Expected: Two routes: local, and 0.0.0.0/0 to an id beginning nat-.

Write both times in your ledger. The address and the gateway bill from now.

Use the window

6

On `thara-web-1`, reach the internet, see which address it sees, and patch.

Expected: An HTTP status line; your Elastic IP; then Complete! or Nothing to do.

```
$ curl -m 10 -I https://aws.amazon.com
$ curl -s https://checkip.amazonaws.com
$ sudo dnf clean metadata
$ sudo dnf -y update
```

Close the window

7

Remove the 0.0.0.0/0 route. Delete the NAT gateway and wait for Deleted. Release the Elastic IP. Then run the first curl again.

Expected: No Elastic IPs; the gateway Deleted; one route; a timeout.

In that order: route, gateway, address. Write the elapsed minutes in your ledger.

SEGMENT 2 - 30 MIN



S3 gateway endpoint

Cost: free

The key works, and the bucket is out of reach

8

In IAM, Activate the Day 2 key. On `thara-web-1`, run `aws configure` with the `.csv` values and region `ap-south-1`. List the bucket.

Expected: Connect timeout on endpoint URL.

```
$ aws configure
$ clear
$ aws s3 ls s3://thara-reports-<student id> \
$      --cli-connect-timeout 5
```

Clear the screen before any screenshot. The secret must never appear in one.

Create the endpoint, list the bucket

9

Endpoints, Create endpoint: AWS services, Type Gateway, the S3 service, `thara-vpc`, route table `thara-private-rt` only, Full access.

Expected: Available. A new route in `thara-private-rt`: `pl-` to `vpce-`.

10

List the bucket again, then try the internet again.

Expected: `reports-index.txt` is listed. The `curl` still times out.

```
$ aws s3 ls s3://thara-reports-<student id>
$ curl -m 10 -I https://aws.amazon.com
```

SEGMENT 3 - 55 MIN



Partner VPC and peering

Cost: one extra t3.micro, about USD 0.011 per hour

The partner's network and server

- 11** Create `partnerlab-vpc`, 10.1.0.0/16, and one subnet, 10.1.1.0/24, in zone a. No name for the subnet.
Expected: The subnet uses the VPC's main route table: one route, to local.

- 12** Launch `partnerlab-1`: t3.micro, no key pair, no public IP, a new security group allowing All ICMP - IPv4 from 10.0.0.0/16.
Expected: Running, with a private address 10.1.1.x.

Peer, then route one side

13

Create a peering connection from `thara-vpc` to `partnerlab-vpc`. Accept it. Ping the partner from `thara-web-1`.

Expected: Active, and 100% packet loss.

14

In `thara-private-rt`, add 10.1.0.0/16 with the peering connection as its target. Ping again.

Expected: Still 100% packet loss. Say why before you go on.

```
$ ping -c 3 -W 2 <partner private address>
```

The other side

15

In the main route table of `partnerlab-vpc`, add 10.0.0.0/16 to the peering connection. Ping. Remove that route and ping. Add it back.

Expected: 3 received; then 100% packet loss; then 3 received.

Keep both outputs. The request needs Thara's route. The reply needs the partner's.

SEGMENT 4 - 30 MIN



Ledger, teardown, evidence

Cost: nil

Prove it, then break it



PROVE IT

- Curl and the package update succeeded through NAT; the curl timed out again without the route
- The bucket lists from `thara-web-1` with no default route in `thara-private-rt`
- Ping to 10.1.1.x succeeds only with both peering routes

BREAK IT

- Move the endpoint to `thara-public-rt`:
Manage route tables
- The listing times out, yet the endpoint shows Available
- Find the evidence, say why, restore

Tear down and record



TEARDOWN

- Deactivate the Day 2 key
- Remove the 10.1.0.0/16 route, then delete the peering, `partnerlab-1` and `partnerlab-vpc`
- Keep the endpoint. Stop `thara-bastion` and `thara-web-1`

COST

Estimated credit use: about USD 1.50

FOR YOUR EVIDENCE PACK

- thara-private-rt before and after each of the three changes (NAT route, endpoint, peering route)
- Ledger rows showing NAT gateway created and deleted with elapsed minutes and the elastic IP released
- Ping output with and without the partner-side route

Common mistakes



COMMON MISTAKE

Deleting the NAT gateway releases the Elastic IP.

It does not. The address bills until it is released.



COMMON MISTAKE

One side's route is enough for peering.

Both sides need a route.

Before the next session



READ

IAM User Guide

Policy evaluation logic. Also Shields, chapters 2 and 4.

SELF-CHECK

CLF-C02 bank

Domain 2 set A, 15 items.

PQ4 is due.

TASK

Your group

Agree tier responsibilities and draft a first capstone diagram.

Bring it to the Day 5 lab.

Questions & Discussion



Next: identity, encryption and detection.



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